

From ledger to *forecast*

Scenario analysis and transition planning ask for a different class of data than emissions accounting produces. This briefing sets out what the standards require, what supervisory exercises have already found, and where the first wave of reporters fell short.

IN THIS BRIEFING

9 exhibits · 25 sources · data as at
1 August 2026

STANDARDS

IFRS S2 · AASB S2 · ISSA 5000 /
ASSA 5010 · MAS ENRM

EVIDENCE BASE

ECB · Bank of England · APRA ·
NGFS · ASIC

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WHAT THIS BRIEFING DOES NOT COVER

Reporting thresholds, phase-in dates and deferral timetables. Those are well documented by the regulators themselves and are not the constraint most preparers are actually facing.

HOW TO READ THE EXHIBITS

Each carries its own source line and, where relevant, a note on what is excluded or uncertain. Superscripts resolve to the endnotes. Where a figure rests on a single source, the source is named in the sentence.

The measurement problem is solved. The forecast is not.

Emissions accounting has a defined boundary, a standardised method and an audit trail. Climate scenario analysis and transition planning have none of those things yet — and they are the half of the disclosure that investors, lenders and insurers actually price.

An emissions inventory is a backward-looking statement about an entity's own operations. A resilience assessment is a forward-looking statement about assets, counterparties and capital under futures the entity does not control. The two require different data, and only one of them is currently produced as a matter of course.

That gap is not a preparer failing. Every supervisory exercise run to date has reported the same constraint about its own results, and the FSB and NGFS warned jointly in 2022 that climate scenario analysis exercises **may understate climate exposures and vulnerabilities**.¹¹ The interesting question is not whether the data is missing. It is what the missing data is doing to the disclosures being relied upon.

Six figures.

Three of these describe what supervisors found when they ran the analysis themselves. Three describe what the regime does with the results.

~70%

OF REPORTED SCOPE 1, 2 AND 3 EMISSIONS IN THE ECB'S 2022 STRESS TEST RELIED ON PROXIES RATHER THAN COUNTERPARTY DATA¹²

60%

OF SAMPLED BANKS HAD NO WELL-INTEGRATED CLIMATE RISK STRESS-TESTING FRAMEWORK¹²

Lower

LOSS PROJECTIONS WERE PRODUCED BY CBES FIRMS RELYING ON THIRD-PARTY MODELS THEY COULD NOT CHALLENGE INTERNALLY¹³

0

ASSURANCE OVER SCENARIO ANALYSIS AND TRANSITION PLANS IN THE FIRST AUSTRALIAN REPORTING YEAR²⁰

3 yr

MODIFIED LIABILITY COVERING SCOPE 3, SCENARIO ANALYSIS AND TRANSITION PLANS – THE SAME THREE ITEMS²²

2/3

JUST UNDER TWO-THIRDS OF FIRST-WAVE AUSTRALIAN REPORTERS DISCLOSED A TRANSITION PLAN¹⁸

The least verifiable disclosures are the ones being priced.

IFRS S2 requires an entity to assess the resilience of its strategy and business model using climate-related scenario analysis, and to disclose the current and anticipated financial effects of climate risks on its financial position, performance and cash flows.¹ AASB S2 goes further than the global baseline, requiring at least two scenarios — one consistent with limiting warming to 1.5 °C, one in which warming well exceeds 2 °C.²

Those requirements are met with data that does not yet exist in most organisations. The ECB found roughly **70% of reported emissions rested on proxies**; the Bank of England found value-chain emissions and counterparty transition plans simply unavailable, down to missing postcodes for physical assets; APRA attributed the variance between bank submissions to data quality and the absence of a standardised modelling approach.^{12,13,14}

The regime has, in effect, conceded the point. In Australia, scenario analysis and transition plans receive no assurance in the first reporting year and only limited assurance thereafter,²⁰ and they sit alongside Scope 3 inside a three-year modified liability window.²² Those are the same three disclosures that investors, lenders and insurers use to price transition credibility.

THE CONSEQUENCE

An entity can satisfy the standard and still publish a resilience assessment that nobody — including its own board — can test. ASIC's first review of Australian reports found scenario disclosures lacked detail on the assumptions and dependencies relied on, and transition plans were not clearly linked to targets and actions.¹⁹

01

What the standards ask for

The forward-looking requirements, and the reliefs that sit alongside them. Proportionality is built into the standard — but it is not a permanent exemption.

07 Resilience, and the reliefs

08 Two scenarios, one of them hot

09 What the first wave disclosed

10 A transition plan is a disclosure object

Resilience, and the reliefs.

Exhibit 1

The forward-looking requirements each carry a proportionality relief.

Selected IFRS S2 requirements and the conditions under which quantitative disclosure is not required.

REQUIREMENT	WHAT IS REQUIRED	RELIEF
Climate resilience para 22	Assess the resilience of strategy and business model using scenario analysis; disclose how and when it was carried out, the key assumptions, and significant areas of uncertainty.	Approach must be commensurate with the entity's circumstances, exposure, skills, capabilities and resources.
Anticipated financial effects paras 15–21	Current and anticipated effects on financial position, performance and cash flows, over short, medium and long term.	Quantitative information not required where measurement uncertainty is too high, or where the entity lacks the skills, capabilities or resources.
Transition plan para 14	If the entity has one: key assumptions, dependencies it relies on, and quantitative and qualitative progress against plans disclosed previously.	Conditional — there is no requirement to have a transition plan.
Carbon credits para 36(e)	Extent to which a net target relies on credits, the verifying scheme, the credit type, and factors bearing on credibility and integrity.	None. This sits in targets, not in the transition plan.

Note: reliefs are available to entities of any size. Where exposure to climate-related financial risk is material, the ISSB expects the effort and resources devoted to increase correspondingly — proportionality is not a permanent exemption.

Source: IFRS S2; IFRS Foundation guidance on anticipated financial effects, August 2025

Two scenarios, one of them hot.

IFRS S2 prescribes neither a number of scenarios nor a temperature. AASB S2 prescribes both — the clearest departure from the global baseline, and the one most often mis-stated.

Exhibit 2

Australia mandates a scenario pair; the widely quoted 2.5 °C figure is the regulator’s interpretation, not the standard.

Scenario requirements under IFRS S2 and AASB S2, with the source of each.

INSTRUMENT	WHAT IT SAYS
IFRS S2	Requires scenario analysis for the resilience assessment. Prescribes no number of scenarios and no temperature alignment.
AASB S2 Appendix B, AusB1	At least two scenarios: one consistent with limiting warming to 1.5 °C above pre-industrial levels, and one in which warming well exceeds 2 °C .
ASIC RG 280	Clarifies that a scenario of 2.5 °C or greater satisfies “well exceeds 2 °C”, and that using a lower high-warming scenario creates compliance risk.

Note: the 2.5 °C figure does not appear in AASB S2. It is ASIC regulatory guidance interpreting the standard, and should be attributed accordingly.

Source: AASB S2, Appendix B; ASIC Regulatory Guide 280; ReGenesis Impact analysis

What the first wave actually disclosed.

Exhibit 3

Most first-wave reporters exceeded the scenario minimum, and over half quantified financial effects to some degree.

Selected findings from reviews of Australian Group 1 entities reporting for years ended 31 December 2025.

OBSERVATION	FINDING
Reference pathways	Most entities used established pathways from the IPCC, NGFS or IEA.
Number of scenarios	Over half disclosed more than the required minimum .
Anticipated financial effects	Over half provided some level of quantified information ; approaches ranged from qualitative assessment to advanced quantitative modelling.
Transition plans	Just under two-thirds disclosed one, ranging from operational initiatives to defined investment programmes.
Scope 3	All entities reported Scope 1 and 2; 40% disclosed Scope 3 voluntarily despite first-year relief.
Risk composition	On average 47% transition risk, 31% physical risk, 22% opportunity .

Note: PwC reviewed 22 first-wave Group 1 reporters published on the ASX on or before 27 February 2026. Deloitte’s review covers the same reporting population. First-wave entities represent a small fraction of those eventually captured.

Source: Deloitte, Early insights into Wave 1; PwC Australia, AASB S2 unpacked

A transition plan is a disclosure object, not a pledge.

Under IFRS S2 the plan is disclosed conditionally — but once disclosed it carries assumptions, dependencies and a requirement to report progress against what was said last year.

What must accompany it

Key assumptions used in developing the plan — assumed policy settings, assumed technology availability — and the dependencies the plan relies on. Then, in later periods, quantitative and qualitative progress against what was previously disclosed.¹

Where the guidance sits

The Transition Plan Taskforce materials transferred to the IFRS Foundation in 2024, and in June 2025 it published guidance on disclosing climate-related transition information under IFRS S2. That guidance is **not new requirements** — it does not amend the standard.^{5,6}

THE SUPERVISORY VERSION

MAS took a different route. Rather than a disclosure requirement, its March 2026 guidelines make transition planning a supervisory expectation for banks, insurers and asset managers from September 2027 — requiring scenario results to be integrated into credit, underwriting and investment decisions, and setting expectations on the governance of **proxy data**, including the choice of proxies and its implications for risk assessment.⁷

02

What the exercises found

Four supervisors have now run climate scenario analysis at scale. Each published what it found about the economy. Each also published what it found about its own data.

12 The reference scenario sets

13 A damage function withdrawn

14 What the exercises said about their data

15 Physical risk, priced through insurance

Two scenario sets, two different jobs.

Exhibit 4

The short-term scenarios published in 2025 answer a question the long-term set cannot.

NGFS long-term (Phase V) and short-term scenario families, by horizon and intended use.

	LONG-TERM, PHASE V	SHORT-TERM
Published	November 2024	May 2025
Horizon	To 2050 and 2100	3 to 5 years
Scenarios	Seven pathways, including Net Zero 2050, Low Demand, Below 2 °C, Delayed Transition, NDCs, Current Policies and Fragmented World	Four: Highway to Paris; Sudden Wake-Up Call; Disasters and Policy Stagnation; Diverging Realities
Intended use	Strategic and investment decisions	Capital planning, credit risk, stress testing
Distinguishing feature	Macro-financial pathways by region and sector	Models climate shocks interacting with the economic cycle, capturing acute physical risk and abrupt repricing within a normal business-planning horizon

Note: the older six-scenario framing grouped as Orderly, Disorderly and Hot House World belongs to Phase III–IV and is superseded. “Divergent Net Zero” no longer exists as a Phase V pathway.

Source: NGFS Phase V, November 2024; NGFS short-term scenarios, May 2025

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A damage function withdrawn.

The most consequential development in climate scenario analysis in the last year is not a new scenario. It is the withdrawal of the research underpinning the physical-risk estimates in the set most preparers are using.

NGFS STATEMENT

The academic paper underpinning the physical-risk damage function in Phase V has been **retracted**. The NGFS has issued a formal statement noting that it cannot be excluded that the economic effects of climate change turn out to be more severe than current Phase V estimates. Scenario outputs that do not incorporate those physical loss estimates are unaffected, as are all short-term scenarios. An updated methodology is expected with the next long-term iteration at the end of 2026.^{9,8}

Why it is a disclosure issue

IFRS S2 requires disclosure of the key assumptions and significant areas of uncertainty in the resilience assessment.¹ An assessment built on a withdrawn damage function has a material assumption to report.

What it does not mean

It is not evidence that physical risk is overstated. The NGFS statement points the other way — that effects may prove more severe than Phase V shows.

Every exercise reported the same constraint.

Exhibit 5

Four supervisors ran the analysis and four reported that the data was not there.

Published findings on data quality from supervisory climate scenario exercises.

EXERCISE	WHAT IT REPORTED ABOUT ITS OWN DATA
ECB climate risk stress test, 2022	About 70% of reported Scope 1, 2 and 3 emissions relied on proxies rather than counterparty data. 65% of banks used predominantly proxies to allocate exposures to energy-performance ratings. 60% had no well-integrated climate stress-testing framework. Wide dispersion in estimated borrower Scope 3 — including between data providers for the same counterparty.
Bank of England CBES, 2022	Lack of data on corporates' value-chain emissions and future transition plans was a common issue, down to missing postcodes for physical assets and out-of-date energy-performance ratings. Firms relying on third-party models without the internal capability to challenge them produced materially lower loss projections.
APRA Climate Vulnerability Assessment, 2022	Large variance between bank submissions driven by deviations in data quality and the absence of a standardised modelling approach , described as a still-maturing approach to using scenario analysis.
PwC and IIF survey of financial institutions, 2025	Across 24 institutions managing more than US\$18 trillion , anticipated financial effects were among the most-cited challenges, with respondents pointing to immature internal systems and controls.

Note: the ECB and CBES exercises predate current disclosure requirements; they are included because they are the largest published runs and the only ones reporting data quality in this detail.

Source: ECB, July 2022; Bank of England CBES, May 2022; APRA, November 2022; PwC and IIF, 2025

Physical risk, priced through insurance.

Exhibit 6

Expected weather losses more than double by 2050 under the higher physical-risk scenario.

Expected national annual weather-peril losses, Australia, A\$ billion.



Note: APRA modelled two severe but plausible scenarios to 2050. The 2024 baseline is shown recessive; the projection is the finding.

Source: APRA, Mind the Gap, March 2026

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1 in 4

HOUSEHOLDS IN FREESTANDING PROPERTIES
PROJECTED UNINSURED BY 2050, FROM 1 IN 7
TODAY ¹⁵

1.6_m

AUSTRALIAN HOUSEHOLDS ALREADY IN HOME-
INSURANCE AFFORDABILITY STRESS, UP 30%
YEAR ON YEAR ²⁴

03

The data of the future

What a forward-looking assessment needs, what an emissions inventory produces, and why the difference is not closing on its own.

17 Four properties accounting does not have

18 The assurance sequence

19 Where the regime concedes the point

Four properties an inventory does not have.

Exhibit 7

Forward-looking analysis needs data with a different shape, not simply more of it.

Properties required by scenario analysis and transition planning, against what GHG accounting produces.

PROPERTY	WHAT ACCOUNTING PRODUCES	WHAT THE ASSESSMENT NEEDS
Resolution	Entity-level or facility-level totals	Asset and location level — hazard exposure attaches to a coordinate, not to a legal entity. The CBES failure was missing postcodes, not missing emissions.
Direction	Historical, for a closed period	Conditional on a pathway — the same asset carries different values under 1.5 °C and under well-above-2 °C, and both must be run.
Boundary	Own operations, with Scope 3 estimated	Counterparty and value chain — transition risk sits with customers and suppliers. The ECB's best performers were distinguished by obtaining clients' actual data and transition plans.
Object	Emissions, reported after the fact	Capital, mapped forward — anticipated financial effects are a statement about cash flows, not about tonnes.

Note: the four properties are ReGenesis Impact’s framing of constraints reported separately by the ECB, the Bank of England and APRA; the underlying findings are those supervisors’ own.

Source: ECB, 2022; Bank of England CBES, 2022; APRA, 2022; IFRS S2; ReGenesis Impact analysis

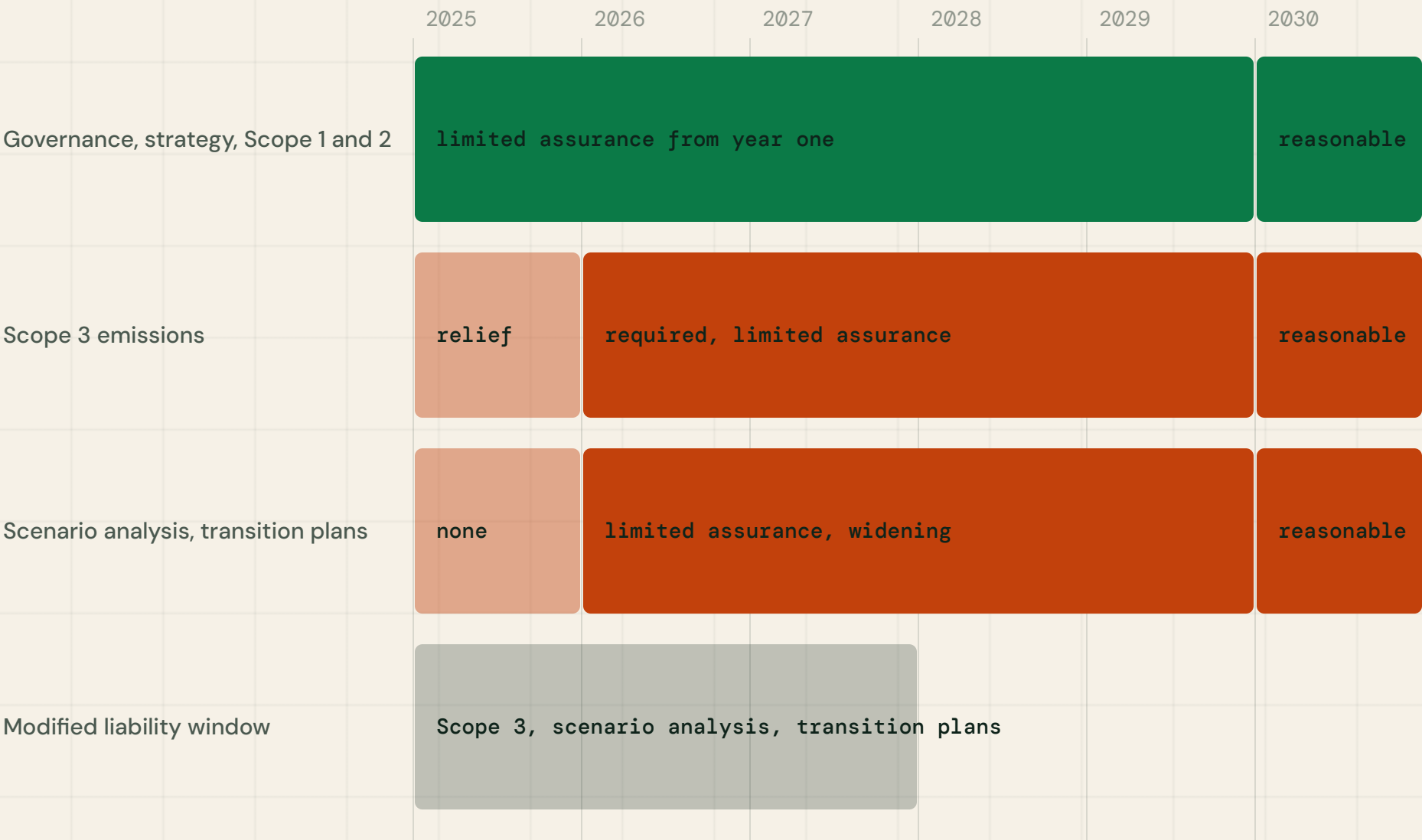
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The assurance arrives last, by design.

Exhibit 8

Scenario analysis and transition plans are the last disclosures to be assured, and the only ones with liability protection.

Australian assurance phasing and the statutory liability window, by disclosure.



Note: reasonable assurance over all climate disclosures applies for years commencing on or after 1 July 2030. The assurance phasing is drawn from ASSA 5010; readers relying on year-one scope should confirm it against the standard.

Source: AUASB, ASSA 5000 and ASSA 5010; Treasury Laws Amendment Act 2024

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ISSA 5000 permits assurance over forward-looking information, while noting it is ordinarily **capable of being evaluated only with less precision** than historical data.²¹

Three disclosures, three concessions, one list.

Scope 3, scenario analysis and transition plans appear together in every concession the regime has made. That is not coincidence — it is the system recording where it knows the evidence is weakest.

Relief

SCOPE 3 NOT REQUIRED IN THE FIRST AUSTRALIAN REPORTING YEAR²

No

assurance

OVER SCENARIO ANALYSIS AND TRANSITION PLANS IN YEAR ONE²⁰

3 yr

MODIFIED LIABILITY COVERING ALL THREE²²

ASIC's first review of Australian reports named the two weaknesses directly: scenario analysis disclosures **lacked detail about the underlying assumptions and dependencies relied on**, and transition plans **could have been more clearly linked to the entity's targets, actions and strategies**. The same review recorded a marked improvement in the quantity and quality of climate information overall.¹⁹

WHAT FOLLOWS FROM THAT

The reliefs close. Assurance widens to the full report from year two and becomes reasonable from July 2030; the liability window is three years.^{20,22} The disclosures currently carrying the least scrutiny are the ones that will carry the most.

04

Why stakeholders read it

Transition credibility is already priced in loan spreads, underwriting and coverage decisions. The disclosure is read because something turns on it.

Who is reading, and what turns on it.

Exhibit 9

Each reader group has a decision attached to the disclosure, and a stated view of its reliability.

Published evidence on the use of transition plans and scenario analysis, by reader.

READER	WHAT THEY DO WITH IT	STATED POSITION
Investors	Assess whether sustainability risk management bears on the investment case.	94% believed corporate sustainability reporting contains unsupported claims; 75% said sustainability risk management was an important factor in investment decisions. ²³
Lenders	Score counterparty transition maturity as a condition of transition finance.	Deutsche Bank operates a Transition Maturity Score on a 1–7 scale, assessing counterparty plans across transition management and performance. ²⁵
Supervisors	Require scenario results to be integrated into credit, underwriting and investment decisions.	MAS guidelines effective September 2027 for banks, insurers and asset managers. ⁷
Insurers	Price and, at the margin, withdraw cover.	The Australian home-insurance protection gap is projected to widen from one in seven households to one in four by 2050. ¹⁵

Note: the investor figures are from different editions of the same survey and are not a time series. The Deutsche Bank framework is one lender’s published methodology, cited as a concrete example rather than as market practice.

Source: PwC Global Investor Survey; Deutsche Bank Transition Finance Framework; MAS; APRA

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Five actions.

- **Record the scenario assumptions as a governed artefact, not a paragraph.** ASIC's stated weakness was missing detail on assumptions and dependencies — which is a documentation problem before it is an analysis problem.^{19,1}
- **Establish asset-level location data before the next assessment cycle.** The CBES constraint was missing postcodes. Physical hazard exposure cannot be modelled against a registered office.¹³
- **Treat counterparty transition data as a collection programme.** The ECB's better performers were distinguished by engaging clients for actual data and their transition plans, not by better modelling.¹²
- **State which scenario vintage was used, and when.** Phase V physical-risk estimates rest on a withdrawn damage function; an assessment that does not say what it ran cannot be re-run.⁹
- **Build the evidence trail for the disclosures that are not yet assured.** Scenario analysis and transition plans move from no assurance to limited, then to reasonable by July 2030, and the liability window is three years.^{20,22}

THE ASYMMETRY

Each of these is a data-collection decision with a lead time measured in reporting cycles, not weeks. An organisation that begins after the requirement bites will be assured on a baseline it did not have time to build.

Scope, method and limitations.

Scope. Forward-looking climate disclosure requirements — resilience assessment, scenario analysis, anticipated financial effects and transition planning — under IFRS S2, AASB S2 and the MAS transition planning guidelines, together with the published supervisory exercises that have tested them. Reporting thresholds and phase-in timetables are deliberately out of scope.

Method. Compiled from public standards, regulatory guidance, supervisory publications and published reviews of first-wave reporters, listed in the endnotes. Where a figure rests on a single published source, that source is named in the sentence rather than only in the endnote. Where two figures measure different populations or periods, they are not combined. **Data as at 1 August 2026.**

LIMITATIONS

Three should be read alongside the figures. This is a compilation of published positions, not a survey — it describes what supervisors and reviewers reported, not what companies did. The Australian first-wave findings cover 22 to 25 entities reporting for years ended 31 December 2025, a small fraction of those eventually captured, and should not be read as settled practice. And the assurance phasing described on page 18 is drawn from secondary summaries of ASSA 5010; the scope of first-year assurance should be confirmed against the standard before it is relied on.

Sources.

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Figures are quoted as published by the issuing body. Claims that could not be corroborated across independent sources have been excluded rather than qualified; one figure previously carried in this series was withdrawn after the underlying paper was retracted.

Who produced this briefing.

ReGenesis Impact builds disclosure tooling for companies in Australia and Singapore — the GHG inventory, the ISSB and AASB disclosure structures, scenario analysis and the assurance evidence trail described in this briefing. The tools are free to use and require no account to start.

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Corrections are welcome. If a requirement, figure or date in this briefing has moved or has been misread, please write and it will be amended in the next edition.

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